DreaMS-conditioned Hybrid MS2LDA: Pretrained Spectral Conditioning for Neural LDA Inference and Mass2Motif Discovery

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Abstract

MS2LDA discovers recurring fragment and neutral-loss patterns by treating tandem mass spectra as documents in latent Dirichlet allocation (LDA). This note specifies a hybrid in which a frozen pretrained DreaMS transformer conditions both amortized document inference and a learned, fixed-mass Dirichlet prior, while classical expected-count updates continue to discover free topics. The prior is learned for a bounded number of epochs and then frozen so convergence is assessed against a stationary model. The reference implementation exposes a small Tomotopy-shaped interface without changing the production MS2LDA backend. It is accompanied by focused correctness tests and a prespecified validation protocol. No comparative performance or chemical motif-quality claim is made here; those claims require a new reproducible benchmark fitted directly from this cleaned implementation.

1 Introduction

Latent Dirichlet allocation represents a document as a mixture of latent topics and each topic as a probability distribution over words [1]. MS2LDA transfers this construction to tandem mass spectrometry: spectra are documents, discretized fragments and neutral losses are words, and topics are Mass2Motifs that may represent recurring molecular substructures [3]. One spectrum can contain several motifs, and one motif can recur across otherwise different molecules.

There are two distinct tasks. In *discovery*, the topic–word distributions are learned from a corpus. In *decomposition*, a new spectrum is assigned mixture weights over already learned topics. A network that predicts membership in fixed motifs can accelerate decomposition, but it cannot discover new motifs. A fully neural topic model can learn both parts, but changes the optimizer and can suffer component collapse or unstable topic identification [4].

This reference takes a middle route. Free topics are updated from variational LDA expected counts, while a frozen public spectral encoder supplies continuous features to an empirical-Bayes word prior and an amortized new-document initializer. The scientific question is whether these features add useful chemical structure without replacing LDA's auditable topic update. That question remains to be answered by the planned benchmark; this document first defines exactly what will be tested.

The contributions are:

1. a frozen DreaMS feature extractor that returns aligned spectrum-level and contextual peak representations;

2. one hybrid LDA model that uses those representations in amortized document inference and a structured topic–word prior;

3. a deliberately small adapter surface for fitting, inference, topic access, and safe inference-only checkpoints; and

4. a reproducible protocol for later comparison with Tomotopy and the frozen DreaMS representation itself.

2 Background

2.1 Classical and neural topic models

Mean-field variational inference for LDA alternates local document updates with global expected-count updates [1, 2]. Tomotopy uses a collapsed Gibbs sampler rather than mean-field variational inference, but fits the same broad LDA factorization. Both methods are stochastic and label-invariant: different seeds can find different local solutions even when all topics are used.

Neural topic models such as ProdLDA replace repeated local optimization with an amortized encoder [4]. This makes inference fast and allows continuous covariates, but creates an amortization gap: one shared mapping need not return the best variational posterior for every document [5]. Semi-amortized inference reduces this gap by using the encoder output as the start of local optimization [6]. Our model uses this idea while retaining a conjugate LDA topic posterior.

2.2 Pretrained DreaMS representations

DreaMS is a transformer pretrained by self-supervision on millions of unannotated tandem mass spectra [7]. It emits 1,024-dimensional contextual representations for spectral peaks and a spectrum-level representation at its precursor token. Its training objective and attention architecture encode peak context and pairwise mass differences, information absent from an MS2LDA count vector. We use the public `DreaMS_embedding` checkpoint without fine-tuning the transformer.

3 Method

3.1 LDA core

Let x_{dv} be the count of feature v in spectrum d, with K motifs and V retained features. Conditional on the empirically estimated $\boldsymbol{\eta}$, the model is classical LDA:

$$\begin{aligned}
\boldsymbol{\beta}_k &\sim \mathrm{Dirichlet}(\boldsymbol{\eta}_k), && (1)\\
\boldsymbol{\theta}_d &\sim \mathrm{Dirichlet}(\boldsymbol{\alpha}), && (2)\\
z_{dn} &\sim \mathrm{Categorical}(\boldsymbol{\theta}_d), && (3)\\
w_{dn} &\sim \mathrm{Categorical}(\boldsymbol{\beta}_{z_{dn}}). && (4)
\end{aligned}$$

The topic–word variational posterior is $q(\boldsymbol{\beta}_k) = \mathrm{Dirichlet}(\boldsymbol{\lambda}_k)$, and the document posterior is $q(\boldsymbol{\theta}_d) = \mathrm{Dirichlet}(\boldsymbol{\gamma}_d)$. For each nonzero feature count,

$$\phi_{dvk} \propto \exp\left[\psi(\gamma_{dk}) - \psi\left(\sum_j \gamma_{dj}\right) + \psi(\lambda_{kv}) - \psi\left(\sum_u \lambda_{ku}\right)\right], \tag{5}$$

$$\gamma_{dk} = \alpha_k + \sum_v x_{dv}\phi_{dvk}, \tag{6}$$

$$\lambda_{kv} = \eta_{kv} + \sum_d x_{dv}\phi_{dvk}. \tag{7}$$

The last equation is a full-corpus expected-count update. The neural components do not replace it.

The implementation names the sparse count matrix `x`, the local parameters `gamma`, the responsibilities `phi`, and the global parameters `lambda_posterior`. The method `beta_mean()` returns the posterior mean of $\boldsymbol{\beta}$. These names intentionally follow the equations.

3.2 Frozen spectrum and peak features

For spectrum d, the frozen DreaMS checkpoint produces a global representation $\mathbf{s}_d \in \mathbb{R}^{1024}$ and a contextual representation $\mathbf{h}_{dp} \in \mathbb{R}^{1024}$ for each retained peak p. The extractor uses the official DreaMS preprocessor and its top 100 peaks.

Contextual word features are constructed from training spectra only. For a fragment word at mass m_v, we pool the state of the nearest peak within 0.02 Da. For a neutral-loss word at loss ℓ_v, we pool the nearest peak to precursor mass minus ℓ_v. Repeated occurrences contribute according to their counts. This gives a mean contextual vector $\bar{\mathbf{h}}_v$. Unmatched words receive a zero contextual vector but retain their mass and word-type features.

3.3 Chemically structured topic–word prior

The contextual vector, numeric mass, and feature type are combined into a trainable normalized word representation:

$$\mathbf{u}_v = \mathrm{normalize}\left[W_h\bar{\mathbf{h}}_v + g_m(\widetilde{m}_v) + \mathbf{e}_{\mathrm{type}(v)}\right]. \tag{8}$$

The scalar input is $\widetilde{m}_v = \log(1+\max(m_v,0))/\log(2001)$. Here g_m is a small multilayer perceptron, and $\mathbf{e}_{\mathrm{type}(v)}$ distinguishes fragments, losses, and other tokens. Each topic has a trainable normalized vector $\mathbf{t}_k$. The prior for topic k is

$$p_{kv} = \mathrm{softmax}_v\left(\frac{\mathbf{t}_k^\top \mathbf{u}_v}{T}\right), \tag{9}$$

$$\eta_{kv}^{(e)} = \eta + r_e\,\rho\,\frac{N}{K}p_{kv}, \tag{10}$$

where N is the number of training tokens, $T = 0.5$, $\rho = 0.05$, and $r_e = \min(e/15, 1)$ is a 15-epoch warm-up. Every topic therefore receives the same structured prior mass.

With row-normalized topic vectors collected in $\widetilde{T}$, the prior parameters minimize

$$-\frac{1}{KV}\sum_k \mathbb{E}_q[\log \mathrm{Dir}(\boldsymbol{\beta}_k; \boldsymbol{\eta}_k^{(e)})] + 10^{-3}\frac{\|\widetilde{T}\widetilde{T}^\top - I\|_F^2}{K^2}. \tag{11}$$

One Adam update is taken per global epoch through epoch 20. The prior is then fixed. A run stops only after the relative L_1 change in $\boldsymbol{\lambda}$ is below 10^{-3} for three consecutive fixed-prior epochs, with a maximum of 100 epochs. This is an operational stopping rule, not a proof of ELBO convergence.

3.4 DreaMS-conditioned document encoder

The encoder is anchored to the current free topics. Define

$$\ell_{kv} = \psi(\lambda_{kv}) - \psi\left(\sum_u \lambda_{ku}\right), \tag{12}$$

$$r_{vk} = \mathrm{softmax}_k(\ell_{kv}), \tag{13}$$

$$e_{dk} = \sum_v \frac{x_{dv}}{N_d} r_{vk}, \qquad N_d = \sum_v x_{dv}. \tag{14}$$

The spectrum representation is projected to 128 dimensions and concatenated with $\mathbf{e}_d$. Two 256-unit hidden layers followed by an output layer predict a residual:

$$\mathbf{a}_d = g_\omega \left[\mathbf{e}_d; \mathrm{proj}(\mathbf{s}_d)\right], \tag{15}$$
$$\widehat{\boldsymbol{\pi}}_d = \mathrm{softmax}\left(\log(\mathbf{e}_d + \epsilon) + \mathbf{a}_d\right), \tag{16}$$
$$\boldsymbol{\gamma}_d^{(0)} = \boldsymbol{\alpha} + N_d \widehat{\boldsymbol{\pi}}_d. \tag{17}$$

The final residual layer starts at zero, so the untrained network begins from topic evidence supplied by LDA rather than an unrelated coordinate system. The encoder is trained by $\mathrm{KL}(\boldsymbol{\theta}_d^* \| \widehat{\boldsymbol{\theta}}_d)$ against stop-gradient locally refined posteriors.

Training VB starts from the retained $\boldsymbol{\gamma}_d$ from the preceding epoch, not from the encoder. Consequently, global spectrum embeddings affect new-document inference but not discovered topics. Only pooled contextual peak features affect discovery, through the structured word prior. This is an alternating amortizer, not a ProdLDA-style end-to-end ELBO model.

During discovery, document posteriors are refined with up to 50 sparse local coordinate updates and tolerance 10^{-4} before the global expected-count update. For a new spectrum, the encoder returns an immediate estimate, which may be followed by standard local LDA updates. Refinement changes only that document's topic mixture; it does not change learned motifs.

3.5 Training and inference sequence

The implementation follows one explicit sequence:

1. build a training-only vocabulary and sparse document–word matrix, then initialize the topic posterior from a seeded random simplex;
2. refine each training document's $\boldsymbol{\gamma}_d$ and $\boldsymbol{\phi}_{dv}$ using the current topics;
3. train the document encoder to reproduce those refined local posteriors;
4. accumulate full-corpus expected topic–word counts, update the bounded structured prior during its declared window, and set $\boldsymbol{\lambda}$ to prior plus expected counts;
5. stop only under the stationary-prior convergence rule; and
6. for an unseen spectrum, compute the encoder initialization and optionally apply local VB without changing $\boldsymbol{\lambda}$.

No dense document-by-vocabulary tensor is constructed. Each batch contains only padded nonzero word identifiers and counts.

3.6 Implementation and compatibility boundary

Conceptually the system is one fitted `HybridLDAModel` plus one frozen external feature extractor. `MS2LDA/hybrid_lda.py` contains the sparse LDA equations, neural modules, global update, checkpoint, and adapter. `MS2LDA/dreams_features.py` contains the lazy DreaMS loader, aligned HDF5 cache, and train-only peak-to-word pooling. There is no second model implementation.

`HybridLDAModel` implements only the small subset of `tomotopy.LDAModel` needed by model-facing experiments: adding documents, training, topic access, new-document inference, document topic distributions, likelihood summaries, and save/load. It is not wired into the existing CLI, dashboard, or visualizer. Tomotopy remains the production backend.

The reference checkpoint contains the topic posterior, neural parameters, configuration, and vocabulary. It deliberately excludes training documents, optimizer state, and random-number-generator state and is therefore inference-only. It is loaded with PyTorch's weights-only mode. Feature provenance is stored in the DreaMS cache rather than the model checkpoint, so callers must keep compatible artifacts together.

4 Validation status

The current evidence is limited to implementation-level checks. The focused tests verify configuration invariants, sparse count handling, the two local LDA equations, conservation of token and Dirichlet mass, fixed prior mass, stationary-prior convergence, deterministic same-seed training, amortized inference, topic and document accessors, safe checkpoint round trips, feature caching, peak-to-word pooling, and the shared spectral-word parser.

These tests show that the code implements the stated algorithm. They do not show that its learned topics are chemically meaningful, more predictive, more stable, or faster than another method. Comparative claims will be added only when the clean implementation is retrained by the protocol below.

5 Planned comparative benchmark

5.1 Artifacts and split discipline

The benchmark will live outside the reference implementation on a validation branch. It must contain a single documented driver, a redistributable input or an immutable input manifest, exact train/test assignments, configuration and seed records, and machine-readable outputs. Generated feature caches and model checkpoints may remain external when large, but their hashes and construction commands must be recorded.

Vocabulary construction and pooled word features must use training spectra only. Two 80/20 evaluations will be frozen before results are inspected:

1. a seeded random spectrum split; and

2. a grouped split that keeps DreaMS-similar spectra on the same side.

For document completion, fragment and neutral-loss words originating from the same physical peak must be kept together. A test-spectrum DreaMS embedding must be recomputed from its observed peak groups rather than from the complete spectrum.

5.2 Compared representations

The primary topic-model comparison is the cleaned hybrid against the existing Tomotopy LDA backend, trained from scratch with the same topic count and input counts over seeds 42–46. A raw global-DreaMS similarity baseline is also required. It is not a competing topic model; it tests whether mapping the pretrained representation through topics adds information beyond using that representation directly.

Training budgets and inference settings must be stated explicitly. Report both model-side latency with cached DreaMS features and end-to-end latency that includes feature extraction. A latency-matched Tomotopy setting should accompany its standard inference setting.

5.3 Metrics and claims

Predictive quality will use held-out negative log likelihood per token:

$$\mathrm{NLL} = -\frac{1}{N_{\mathrm{held}}} \sum_{d,v} x_{dv}^{\mathrm{held}} \log \left(\sum_{k} \widehat{\theta}_{dk} \widehat{\beta}_{kv} \right). \tag{18}$$

Lower NLL is better. Topic diagnostics will include active topics, top-word diversity, word-co-occurrence NPMI, and cross-seed stability after optimal topic matching. Same-seed matched cosine and top-word Jaccard similarity will describe overlap between methods, not establish correctness.

Chemical evaluation must use labels withheld from training and model selection. The preferred primary endpoint is recovery of existing manual motif–spectrum annotations on an independent, redistributable corpus. Spectral-family retrieval may be reported as supporting evidence, with family-macro average precision and paired uncertainty intervals, but spectral network components are not direct substructure annotations.

Metrics, exclusions, seed count, and the primary endpoint must be fixed before test results are inspected. All seeds must be reported. Results from this protocol should replace the validation-status section rather than be combined with exploratory numbers from older implementations.

6 Limitations

The hybrid retains LDA's non-convex, label-invariant topic discovery, so different seeds can still reach different solutions. DreaMS is trained to represent whole-spectrum molecular context, whereas a Mass2Motif is intended to capture a smaller recurring pattern. A useful global representation does not therefore guarantee a useful motif prior.

The structured-prior mass, warm-up, and temperature are fixed design choices that have not yet been validated. The document encoder is trained after each local update and therefore trails the newest global topic update by one epoch; this difference should be small only near convergence. The model is not an end-to-end neural topic model, and the production MS2LDA workflow does not yet select it as a backend. Finally, a passing software test suite cannot replace external chemical validation.

7 Reproducibility

The self-contained reference surface consists of `MS2LDA/hybrid_lda.py`, `MS2LDA/dreams_features.py`, `tests/test_hybrid_lda.py`, `environment-hybrid.yml`, and this document. A small lazy import in `MS2LDA/__init__.py` prevents the full application stack from loading when the reference modules are imported.

The environment uses Python 3.11 and pins NumPy 1.25.0, PyTorch 2.2.1, and DreaMS at Git commit `dbec3a0b514a99e5056cfccde4559fda8cfe8129`. The required DreaMS legacy architecture package is pinned at `d35b34b0caee38348098cda852bff30d8b25cd25`. Installing DreaMS with `--no-deps` is deliberate: the YAML supplies the packages imported by the embedding path and prevents DreaMS's moving `msml@main` requirement from replacing the exact legacy-architecture commit.

The reference can be installed and tested with:

```
conda env create -f environment-hybrid.yml
conda activate ms2lda-hybrid
python -m pip install --no-deps \
  "git+https://github.com/pluskal-lab/DreaMS.git@dbec3a0b514a99e5056cfccde4559fda8cfe8129"
python -m pip install --no-deps -e .
python -m pytest -q tests/test_hybrid_lda.py
```

DreaMS caches record the expected package commit, installed package version, both public checkpoint hashes, embedding dimension, and peak limit. Model loading does not enforce a cache match. Reference checkpoints contain no spectra and cannot resume training.

8 Conclusion

DreaMS-conditioned Hybrid MS2LDA is a focused reference implementation of conjugate free-topic discovery with two neural additions: a DreaMS-conditioned topic–word prior and an amortized new-document initializer. The mathematical ownership is explicit: expected counts discover topics, the structured prior biases their word distributions, and the encoder accelerates document inference. The reference is ready for a clean benchmark, but it does not yet justify replacing Tomotopy or claiming chemically superior Mass2Motifs.

References

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